

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

SMOKING CESSATION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADHD Attention Deficit Hyperactivity Disorder

BP Blood Pressure

CPD Cigarettes Per Day

CAD Coronary Artery Disease

CO Carbon Monoxide

CYP Cytochrome P450

DSM-5 Diagnostic and Statistical Manual of Mental Disorders

HR Heart Rate

NHP Natural Health Product

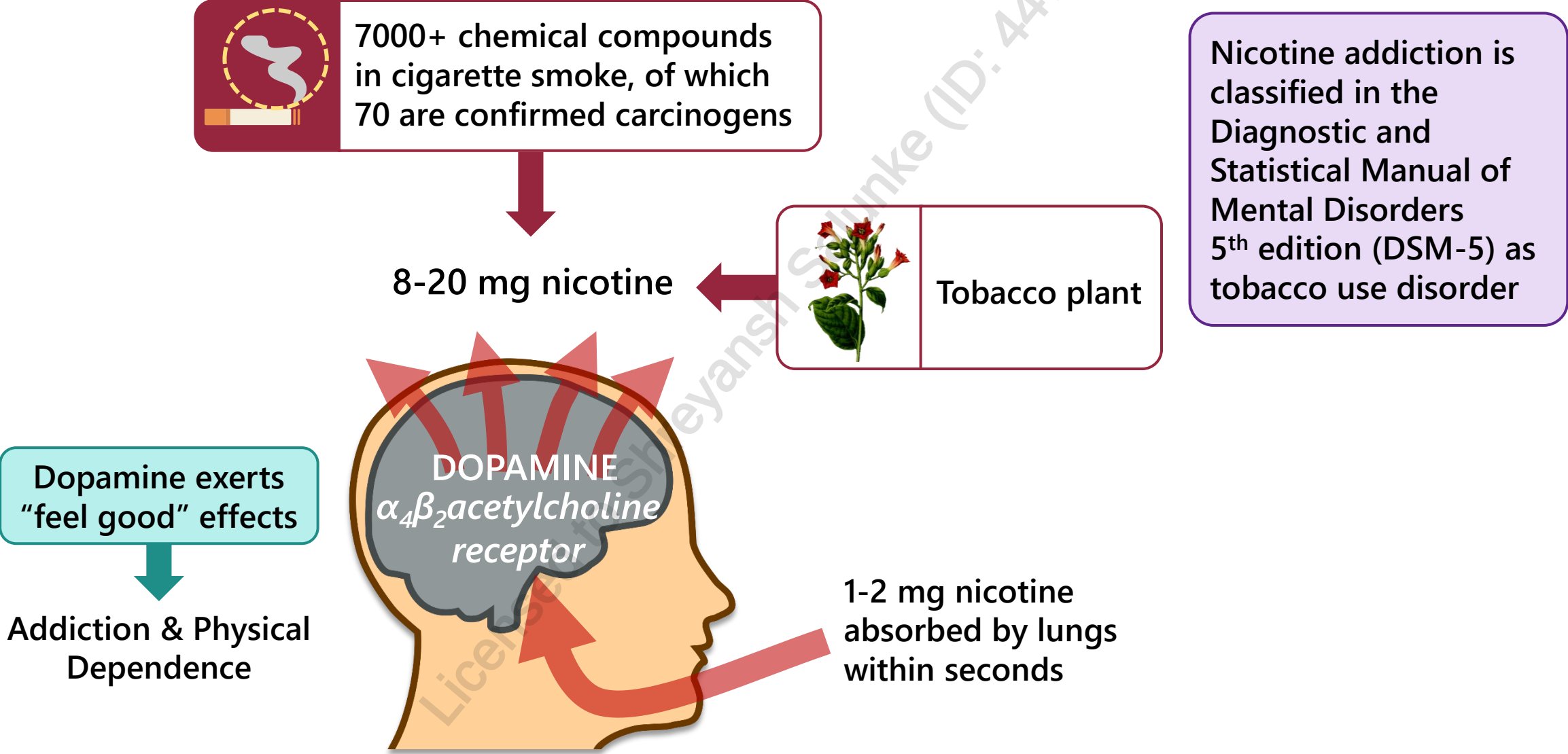
NRT Nicotine Replacement Therapy

WHY SMOKING CESSATION?

Smoking is the #1 cause of preventable disease and death in Canada

Smoking increases the risk of cancer, heart attack, diabetes, stroke, and more

CIGARETTES, TOBACCO, NICOTINE, DOPAMINE



NICOTINE

Beneficial Effects

- Improves anxiety, depression, schizophrenia
- Decreases appetite
- Improves ADHD, Tourette's, pain
- Triggers alertness and memory

Harmful Effects

- Addictive
- Teratogenic
- Increases blood pressure (5-10 mmHg)
- Increases heart rate (10-20 bpm)

Nicotine Withdrawal

Patient
smokes
cigarette

Nicotine
releases
dopamine

Patient feels
temporarily
energized,
happy, alert, or
calm

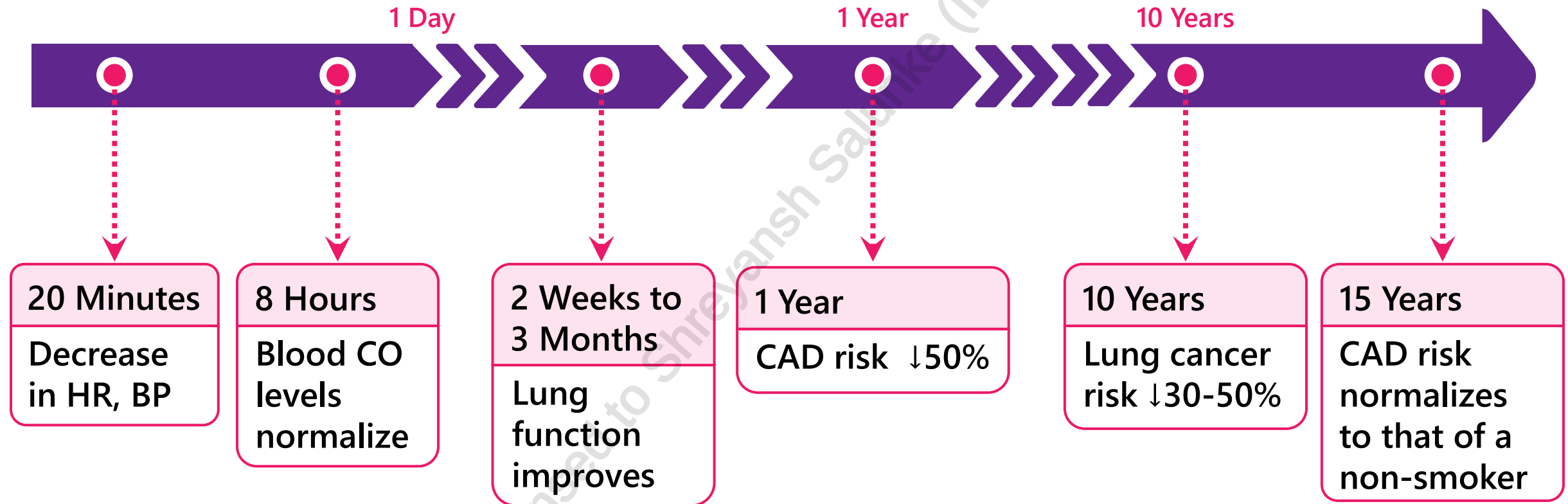
Nicotine levels
decrease

Brain and body
crave nicotine

- Feelings of
discomfort
and irritability
persist

Patient
smokes
cigarette

BENEFITS OF QUITTING



5 A'S

Systematic approach to treat tobacco use and dependence:



ASK

What is your patient's smoking status?

When was the last time you used any tobacco products?

Since tobacco-use can impact your health and medications, do you use any tobacco products?

Advise

Highlight the importance of smoking cessation in a manner that is personally relevant

I can't believe I had a heart attack last year...

I want to be able to run with my kids without getting out of breath...

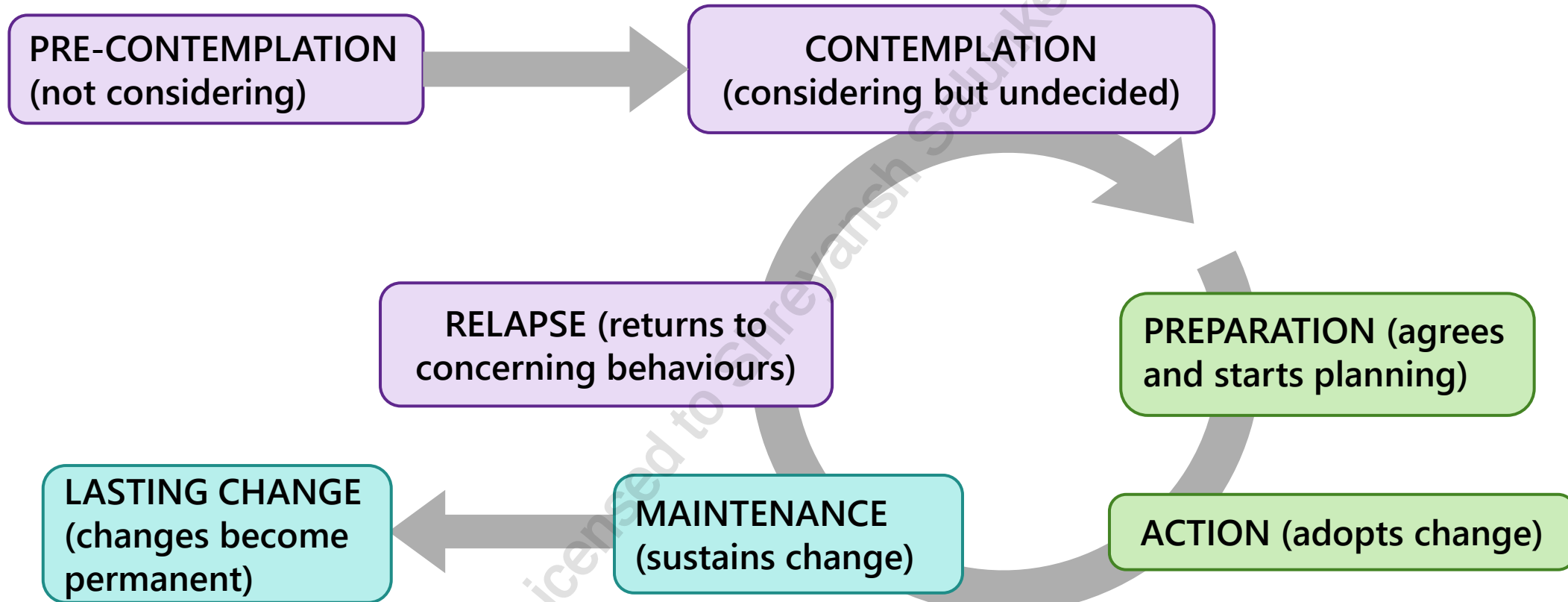
Assess

Is your patient willing to quit? Is your patient ready for change?

Are you thinking about quitting smoking?

THE CYCLE OF CHANGE

Before recommending any behavioural changes, clinicians should assess patients' readiness, willingness, and barriers to change



THE CHANGE SPECTRUM

0

10

RESISTANT



E.g. "I don't want to quit smoking, leave me alone."

AMBIVALENT



E.g. "I want to quit smoking BUT I heard that nicotine patches are expensive."

READY



E.g. "I'm ready to quit smoking."

Assist

Create a quit plan with your patient, recommend appropriate pharmacological therapy, and offer educational resources

Let's create a quit plan that best works for you.

ASSIST

Create a **quit plan** with your patient, recommend the appropriate pharmacological therapy, and offer educational resources:

1



Set a realistic quit date

2



Anticipate barriers:

- Identify situations that trigger smoking habits
- Educate patients regarding withdrawal symptoms: irritability, lack of focus, hunger, headaches, flu-like symptoms. Effects peak 2-3 days after quitting and decrease over weeks-months

3



Enlist social support

5 D's to reduce Cravings

- Delay
- Distract
- Deep Breathing
- Drink Water
- Discuss

ASSIST - What are the available Pharmacological options?

APPROVED PHARMACOTHERAPY OPTIONS:

- ✓ Nicotine Replacement Therapy (NRT): Patches, Gum, Lozenges, Inhaler, Spray
- ✓ Varenicline
- ✓ Bupropion sustained-release
- ✓ Cytisine – *Natural Health Product*

Alternatives:

- ✓ Nortriptyline (not approved by Health Canada for smoking cessation)

RECOMMENDED AGAINST:

- ✓ Acupuncture
 - ✓ Hypnotherapy
 - ✓ Electrostimulation
 - ✓ St. Johns wort
 - ✓ Electronic cigarettes
- (among others)

Create a quit plan with your patient, recommend appropriate pharmacological therapy, and offer educational resources.

Smoking Status	<10 CPD	10-29 CPD	≥30 CPD
Patch	14 mg	21 mg	28 mg (21 mg + 7 mg)

Duration of NRT:
Best results if used for at least 8-12 weeks

Smoking Status After Follow Up	0 CPD	≤5 CPD	6-9 CPD	≥10 CPD
Patch	Continue current dose	Add 7 mg to current dose	Add 14 mg to current dose	Add 21 mg to current dose

When initiating a patch, select an additional short-acting NRT for breakthrough cravings (e.g. nicotine gum 1 piece every hour as needed)

When the patient is ready to reduce NRT:

- Reduce patch by 7 mg q1-2weeks until off patch, then reduce short-acting NRT until not needed

NRT PATCH

NRT

VARENICLINE

BUPROPION

Properties	Consistent nicotine delivery
Strength	7 mg, 14 mg, 21 mg
Adverse effects	• Skin reactions • Insomnia • Vivid dreams
Administration	• 1 patch/day (24-hour) OR • 16-hour application (remove at bedtime) • Apply to dry hairless area of upper arm, upper chest, or hip • Rotate application site
Time to peak serum levels	2-6 hours
Comments	Remove patch before prolonged, strenuous exercise; re-apply after exercise

NRT GUM

NRT

VARENICLINE

BUPROPION

Properties	Rapid relief of cravings; buccal absorption
Strength	2 mg, 4 mg
Adverse effects	• Jaw soreness • Hiccups • Nausea • Heartburn • Headache
Administration	• Flexible dosing • 'Chew and park' technique • Acidic foods and fluids impair buccal absorption, avoid for 15 minutes before use
Time to peak serum levels	20-30 minutes

NRT LOZENGE

NRT

VARENICLINE

BUPROPION

Properties	Rapid relief of cravings; buccal absorption
Strength	1 mg, 2 mg, 4 mg
Adverse effects	• Hiccups • Nausea • Headache
Administration	• Flexible dosing • Suck until strong taste and park • Acidic foods and fluids impair buccal absorption, avoid for 15 minutes before use • GI upset if swallowed
Time to peak serum levels	20-60 minutes

NRT INHALER

NRT

VARENICLINE

BUPROPION

Properties	Rapid relief of cravings; helps with hand-to-mouth ritual; <i>buccal absorption</i>
Strength	4 mg nicotine per 10 mg cartridge (where 2 mg is absorbed systemically)
Adverse effects	• Nausea • Headache • Throat irritation
Administration	• Flexible dosing • Short puffs (NOT deep inhalations)
Time to peak serum levels	15 minutes

Adapted from the University of Ottawa heart institute quit plan consult form.

NRT SPRAY

NRT

VARENICLINE

BUPROPION

Properties	Rapid relief of cravings, buccal absorption
Strength	1 mg nicotine/spray
Adverse effects	• Mouth/throat irritation • Headache
Administration	• Flexible dosing: 1-2 sprays every 30-60 minutes • Prime before first use (or after not using for 2 days) • Point nozzle towards inner cheek • Acidic foods and fluids impair buccal absorption, avoid for 15 min before use
Time to peak serum levels	16 minutes

NRT POUCHES

FYI

NRT

VARENICLINE

BUPROPION

- Newest NRT, placed in the mouth between the gum and cheek, allowing for buccal absorption
 - Authorized by Health Canada for use by adults to help quit smoking, available over the pharmacy counter
- Authorized formulation: one pouch = 4 mg of nicotine/ dose, typically recommended in adults who smoke ≥ 25 cigarettes/day

Adverse effects requiring immediate referral to medical provider:	Contraindications:	Symptoms of nicotine overdose:
Swallowing pouch, irregular heartbeat, allergic reaction, dizziness, nausea, vomiting, abdominal pain, severe heartburn, cold sweats, mental confusion, chest pain, leg pain, blurred vision, and/ or oral irritation	<18 years of age, hypersensitivities to nicotine or ingredients (e.g. benzyl alcohol), history of seizures, medical conditions that warrant nicotine avoidance, pregnancy, breastfeeding, concurrent smoking or use of other forms of nicotine while using pouches	Difficulty breathing, marked weakness, hearing or vision disturbances, abdominal pain, cold sweats, dizziness

VARENICLINE

NRT

VARENICLINE

BUPROPION

Properties	Partial agonist of $\alpha_4\beta_2$ acetylcholine receptor
Strength	0.5 mg, 1 mg
Adverse effects	• Vivid/abnormal dreams • Nausea, vomiting • Insomnia • Constipation and flatulence • Headache
Administration	• Start 1-2 weeks prior to quit date • 0.5 mg daily for 3 days, then 0.5 mg BID for 4 days, then 1 mg BID for 3 months
Precautions	• Maximum daily dose in severe renal impairment ($\text{CrCl} < 30 \text{ mL/min}$): 0.5 mg BID • Cardiovascular risk-inconclusive evidence • Caution in patients with history of or existing psychiatric disease, history of suicidal ideation, history of seizures • Lack of data on use in pregnancy

BUPROPION SUSTAINED-RELEASE

NRT

VARENICLINE

BUPROPION

Properties	Norepinephrine <i>dopamine</i> reuptake inhibitor
Strength	150 mg
Adverse effects	• Dry mouth • Weight loss • Seizures • Insomnia • Agitation
Administration	• Start 1-2 weeks prior to quit date • 150 mg daily for 3 days, then increase to BID for 3 months with at least an 8-hour interval between successive doses • Consider maximum daily dose of 150mg/day in patients with CrCl 15-60mL/min
Precautions	• Contraindicated in patients with bulimia, anorexia, or seizures • Substance use disorder or withdrawal • Child-bearing potential, may use in pregnancy if benefits>risks, no good studies on this, may be associated with a small increased risk of congenital malformations • Hepatic or renal failure • Contraindicated in patients taking MAO inhibitors (within 14 days) • Caution in those taking antidepressants, antipsychotics, corticosteroids, stimulants, theophylline, quinolone antibiotics • Strong CYP 2D6 inhibitor, can increase concentrations of atomoxetine, duloxetine, fluoxetine, fluvoxamine, metoclopramide and tamsulosin • Can also decrease efficacy of tamoxifen and codeine by decreasing their active metabolites

ASSIST

COMBINATION THERAPY



Behavioural support is key in combination with pharmacological measures!

More effective and recommended over patch alone

NRT PATCH

NRT GUM, LOZENGE, SPRAY, INHALER

Higher risk of AEs e.g. nausea and sleep disturbances

NRT

VARENICLINE

More evidence needed to confirm if benefit is significant compared to NRT monotherapy

NRT PATCH

BUPROPION

CYTISINE

- Cytisine is a plant-based alkaloid derived from the *Cytisus Laburnum plant* (Leguminosae family) that acts as a partial agonist on the nicotinic acetylcholine ($\alpha_4\beta_2$) receptor
- **Approved by Health Canada as a Natural Health Product (NHP)**; NHPs have less strict regulatory requirements regarding efficacy and safety
- Strength: 1.5 mg
- Available evidence suggests that cytisine is similar in efficacy and may be more effective than NRT; however, poor quality trial data limits its use
- Side effects: sleep disturbances, dyspepsia, nausea
- Complex dosing regimen

NORTRIPTYLINE

- **Not officially indicated** for smoking cessation, but some evidence of efficacy
- Tricyclic antidepressant
- Use for about 12 weeks
- Common adverse effects: sedation, dry mouth, blurred vision, dizziness, tremor, urinary retention, and orthostatic hypotension

SPECIAL POPULATIONS

Pregnancy & breastfeeding	Cardiovascular disease	Chronic Kidney disease	Psychiatric Disorders (suicide ideations)	Epilepsy
Counselling and non-pharmacological options (preferred)	NRT (PATCH)	NRT	NRT	NRT
Intermittent NRT (gum, lozenge etc.) preferred over continuous patch	Varenicline	Varenicline- dose adjustment	Varenicline- Caution in patients with history of or existing psychiatric disease or history of suicidal ideation	Varenicline (use with caution)
Bupropion but may be associated with small increased risk of congenital abnormalities; use if benefits > risks	Bupropion	Bupropion- dose adjustment	Bupropion- Contraindicated in bulimia or anorexia nervosa	

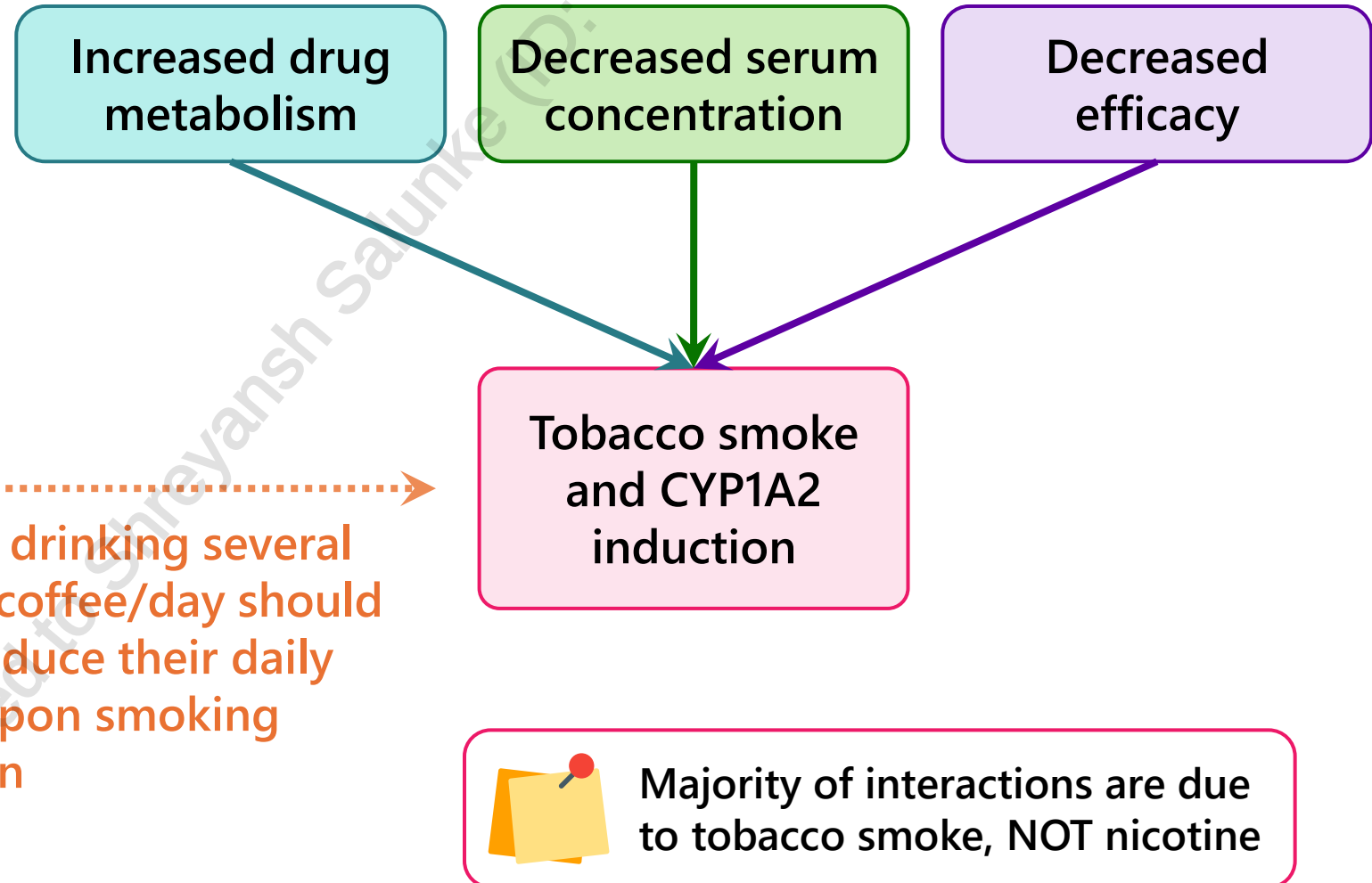
*Patient participation needed to choose best option

DRUG-RELATED CONSIDERATIONS

Create a quit plan with your patient, recommend appropriate pharmacological therapy, and offer educational resources.

- CLOZAPINE
- FLUVOXAMINE
- **CAFFEINE**
- OLANZAPINE
- PROPRANOLOL
- THEOPHYLLINE
- WARFARIN

Patients drinking several cups of coffee/day should halve/reduce their daily intake upon smoking cessation



E-CIGARETTES

Limited studies available

Use is uncertain

May reinforce smoking behaviours

Generally, Not recommended

Arrange

Book a follow-up appointment with your patient

When are you available for a follow-up appointment?

TAKEAWAY

- 5 As: Ask, Advise, Assess, Assist, Arrange
- **APPROVED PHARMACOTHERAPY OPTIONS:**
 - **Nicotine Replacement Therapy (NRT):** Patches, Gum, Lozenges, Inhaler, Spray
 - **Varenicline:** Caution in patients with history of or existing psychiatric disease, history of suicidal ideation, history of seizures
 - **Bupropion sustained-release:** Contraindicated in patients with bulimia, anorexia or seizures, in patients taking MAO inhibitors (within 14 days)
 - **Cytisine:** NHP - similar efficacy/may be more effective than NRT; complex dosing; lack of strong evidence
- Nortriptyline: not officially indicated for smoking cessation

Combination therapy:

- NRT patch + other form of NRT = efficacy
- NRT + varenicline = may be more efficacious than varenicline alone
- NRT + bupropion = limited evidence

CASE

MJ is a 54-year-old construction worker who has smoked 25 cigarettes per day for the past 30 years. MJ has his first cigarette of the day within 1 hour of waking up. He has tried to quit cold-turkey multiple times but has relapsed within weeks each time. MJ shares with you that he would like to quit smoking so he can be active in his children's lives, and he is committed to following a treatment plan this time. His main triggers for smoking include stress and social settings. He was recently diagnosed with hypertension and chronic obstructive pulmonary disease (COPD). His current medications include amlodipine 10 mg daily and salbutamol inhaler 100 mcg 1 to 2 puffs three times daily as needed. He reports mild insomnia but no history of psychiatric illness. His blood pressure today is 130/85 mmHg. He has no allergies to medications.

1. Which phase of the cycle of change is MJ currently in?
 - a) Contemplation
 - b) Action
 - c) Preparation
 - d) Pre-contemplation

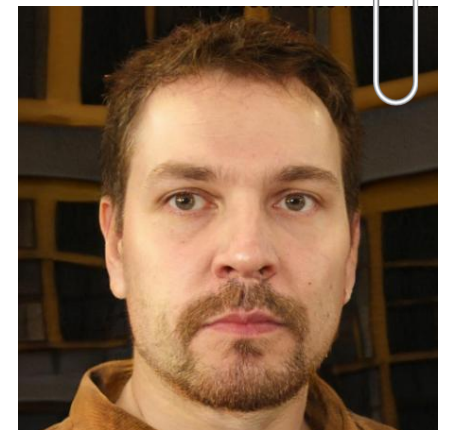


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2. What would be an appropriate starting dose for an NRT patch for MJ?

- a) 7 mg patch
- b) 21 mg patch
- c) 21+7 mg patch
- d) 14 mg patch

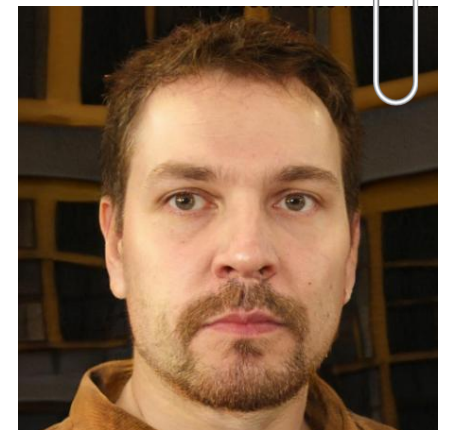


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3. Which of the following is NOT an appropriate counselling point for an NRT patch?

- a) Acidic foods and fluids impair absorption
- b) Remove at bedtime
- c) Rotate application site
- d) Apply to a dry hairless area



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CHANGE LOG

May 2025

- Formatting changes throughout
- Slide 20: Removed 3 mg strength (no longer available)

July 2025

- Some formatting changes made throughout
- Slide 3: Updated legend
- Slide 16: Added "sustained-release" to bupropion
- Slide 17: Modified NRT table
- Slide 23: New slide, currently FYI because not in guidelines, but seen in practice
- Slide 25: Added 'sustained-release' to title

October 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 7: Changed 12 hours to 8
- Slide 16: Updated options as per recently published Canadian guidelines
- Simplified drug tables
- Slide 24: Added caution with the use of varenicline in hx of seizures
- Slides 26 and 27: Expanded to add additional information (minor)
- Slide 30: Minor update in caffeine interaction
- Slide 33: Updated takeaways
- Slide 37: Updated references